

Quick Thermal Performance Estimation of Chip Scale Packaged GaN FETs using a Simple Circuit Model

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Abstract— We present a reduced physics-based thermal model to predict the thermal performance of Chip Scale Package (CSP) GaN FETs mounted to printed circuit boards (PCBs) and provide an easy-to-use thermal characterization tool for engineers and designers to quickly assess thermal management strategies. The heat conductance from the GaN FET to both the PCB side and to the “case” (i.e., die exposed surface) are modeled using resistive-network circuit modeling. The thermal resistance in each conductance path is presented for different parameters relating to PCB construction (stack-up, area, via density), FET size, and cooling approach (TIM material, heat spreader and heatsink). A comprehensive set of thermal finite-element simulations was run over the set of parameters. The results, presented in easy “look-up” style graphs, show that even with a small footprint, GaN FETs power-handling capability can be extended when effectively cooled using a simple PCB design and thermal management approach.

Keywords—GaN FETs, CSP, Thermal modeling,

I. INTRODUCTION

High frequency switching applications that employ gallium nitride (GaN) FETs offer exceptional electrical performance [1] with lower losses compared to equivalent silicon devices. Nevertheless, design engineers must still manage the thermal performance to ensure optimal operation and long-term reliability [2] and be able to extend power capability. Electrical engineers and PCB designers desire a quick characterization tool for initial assessment of the thermal constraints. Among the numerous thermal analysis tools, reduced models [3] are most desired since they are simple to understand and implement, and offer a quick means to determine thermal parameters with an acceptable degree of accuracy.

The presented model implements an interface where designers can define the setup of PCB area and construction (layer stack-up and copper thickness), FET device size and via density, heatsink design, inclusion of a heat-spreader, thermal interface material (TIM), and air-flow rate. These choices can be used to quickly tradeoff various thermal strategies and predict performance. The different thermal design components are represented as thermal resistances and integrated in a circuit model, which is then simplified to report on overall resistance parameters at the case-side and board-side that are more valuable for thermal design.

The model is presented for a single FET and then expanded to consider the widely-used half-bridge (HB) configuration with two FETs in close proximity. The HB configuration and will assess the effect of FET spacing, via density and PCB

construction, as well as general backside cooling parameters. The model can be implemented using general computer analysis tools, such as GNU Octave [4].

II. PATHS OF THERMAL RESISTANCE TO AMBIENT

A. Junction to Board

When using surface mount packaged power semiconductors, the PCB constitutes the main heat conduction path for the mounted devices and is especially critical when no case-side cooling is implemented. Furthermore, packaged device have multiple thermal interfaces between the die and the PCB. In comparison, chip scale packaged (CSP) devices have direct metallic contacts (solder) that offer good electrical and thermal conductance path into the PCB. The heat then spreads into the conductive layers of the PCB where it benefits from the large PCB area (compared to exposed FET area) and dissipates into ambient. Each of the mentioned PCB components introduce some thermal resistance to heat flow and should be modeled for an accurate characterization.

Given their high thermal conductivity, the copper conductors are the main path of heat conduction in the PCB, with the number of layers and weight of copper being the main PCB parameters that determine the PCB’s thermal conductivity. Because of the layered construction, κ_{PCB} is anisotropic with heat conducting more effectively in the in-plane direction. The higher the number of layers, and the thicker the copper, the better the thermal conductance in the in-plane direction (κ_{xy}) which can be estimated using (1) considering a parallel resistance configuration between the copper layers and the dielectric layers (which will be assumed to be FR4).

$$\kappa_{xy, PCB} = \kappa_{Cu} (t_{Cu}/t_{PCB}) + \kappa_{FR4} (t_{FR4}/t_{PCB}) \quad (1)$$

The through-plane thermal conductivity ($\kappa_{z, PCB}$) depends on the series layout of the copper conductor layers and the FR4 layers. Because of this layout, the FR4 becomes the major bottleneck for heat dissipation in the z -direction. One solution is to add thermovias which connect the different conductor layers and offer high conductance paths through the FR4 layer for both electrical current and heat flow. Thermovias reduce the overall thermal resistance and improve the effective $\kappa_{z, PCB}$. The thermal resistance of a single via of

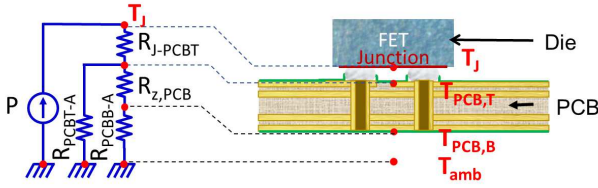


Fig. 1. Thermal resistive circuit showing thermal resistances at PCB side (board-side).

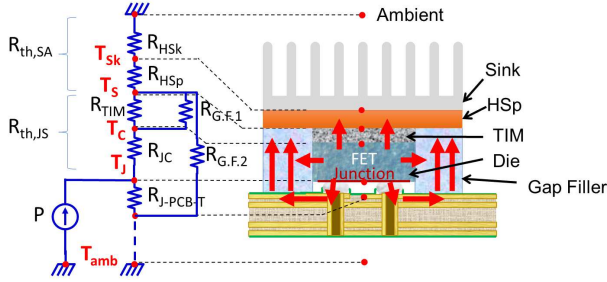


Fig. 2. Thermal resistive circuit showing back-side cooling with TIM pad, gap filler material, a heatspreader and a heatsink

cylindrical construction, and the effective resistance of N number of vias, can be estimated using (2) as function of via diameter (d_{via}), plating thickness ($t_{plating}$), height of PCB (t_{PCB}) and conductivity of copper (κ_{Cu}).

$$R_{th,via} = t_{PCB} / (\kappa_{Cu} \pi d_{via} t_{plating}) \quad (2a)$$

$$R_{th,Nvias} = R_{th,via} / N \quad (2b)$$

The vias' path is perpendicular to the PCB plane, which constitute a higher thermal resistance due to the FR4 layers being in series with the conductor layers. The layered path resistance is calculated using equations (3), and the equivalent through-plane conductivity is given in (4).

$$R_{th,z,PCB} = R_{th,z,FR4} + R_{th,z,Cu} \\ = t_{FR4} / (\kappa_{FR4} A^*) + t_{Cu} / (\kappa_{Cu} A^*) \quad (3)$$

$$R_{th,z} = (R_{th,Nvias}^{-1} + R_{th,z,PCB}^{-1})^{-1} \\ \kappa_{z, PCB} = t_{PCB} / (R_{th,z} A_{PCB}) \quad (4)$$

The heat conducts through the board radially away from the FET region and dissipates into ambient from the surface. The resistance from board to ambient can be estimated assuming radial heat conductance in the PCB and convection heat transfer from the surface with an effective rate h (W/m^2K). The model for circular fin designs has been solved and implemented for a variety of applications.

Since the conductor layers are more clustered near the surface, the effective thermal conductivities differ at the different z locations of the PCB. In the model, three regions

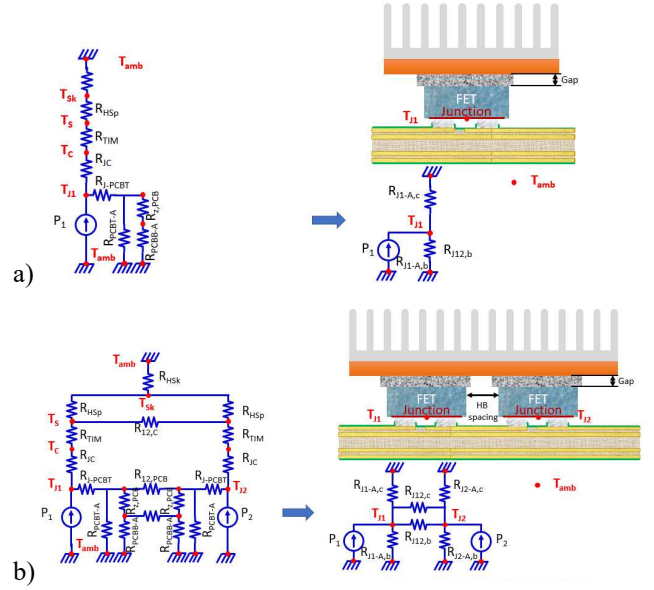


Fig. 3. Schematic of a) Single FET and b) Half-bridge configuration, with full circuit model and circuit model after simplification

are considered for the PCB stack, a top region constituting half of the conductor layers with their FR4 insulation (typically 5 or 10 mils thick), a similar bottom stack with the remaining half of conductor layers and FR4 spacing, and a central region of FR4 core which separates the bottom and top regions and is usually thicker and more insulating (see Fig.1).

The respective contribution of the top and bottom surface to heat dissipation may differ, and thus the two layers (top and bottom) are considered as separate heat conduction paths. Thermal vias serve as thermal connectors to allow heat to conduct from the top stack to the bottom stack. Two temperature nodes are defined for the PCB top and PCB bottom node in the region in the region underneath the FET (Fig. 1) with the respective thermal resistances to ambient (R_{PCBT-A} and R_{PCBB-A}), R_z between the two stacks, and a thermal resistance between the junction and the PCB top (R_{J-PCBT}) to model the effect of the solder bump array and FET construction.

B. Junction to Case

In applications with high power densities, additional cooling may be required to limit self-heating effects by adding a thermal solution to dissipate more heat from the backside (i.e., case-side) of the FETs. The die of CSP GaN FETs is exposed with the silicon substrate having a low junction-to-case thermal resistance ($R_{th,JC}$), in comparison to devices packaged in plastic molds [1-3, 5], and as such can more effectively benefit from back-side cooling.

A simple thermal strategy is to introduce a high conductivity medium (such as copper) for heat to spread and conduct away from the device (see Fig. 2) and reduce hotspots. In addition, heatsinks with protruding fins can be added to further increase the exposed surface area and

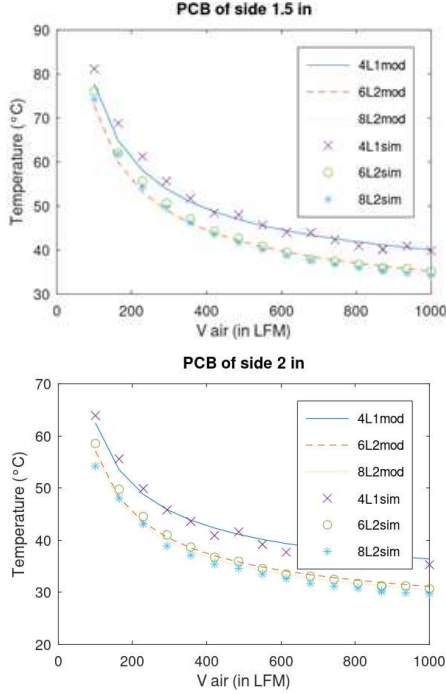


Fig. 4. Graphs showing the variation of PCB temperature as function of air flow and PCB construction. Temperature is extracted from PCB region under FET.

enhance heat dissipation to ambient, making them the go-to solution for quick and easy case-side cooling. To ensure good thermal conductance, thermal-interface materials (TIM) are placed at the contact between the devices and the heatsink bottom surface. TIMs have good thermal properties and come in different forms (pads, gels, putties...) and are selected depending on the application.

The thermal resistance of the TIM material can be estimated from a simple 1D thermal resistance which depends on the material conductivity (κ_{TIM}), the final TIM thickness (t_{TIM}) and the area of the heat conductance from the die (A_{FET}).

$$R_{TIM} = t_{gap} / (\kappa_{TIM} A_{die}) \quad (4)$$

For smaller FETs, the top surface area may be comparable to that of the die sides and as such, heat removal can be enhanced by adding more TIM around the FETs using TIM gel or gap filler. This adds a heat conductance path from the die sides to the sink, and from the top PCB copper to the sink. These paths are represented by their respective thermal resistances R_{GF} shown in the network of Fig. 2.

Beyond the TIM material, heat conducts in the heatspreader and heatsink, which also constitute some resistance to heat conduction. The heat spreading resistance (R_{HSp}) in a conductor block of side s and thickness t_{sp} can be estimated using the reduced model equations presented by Lee et al. [6], while a reduced model for estimating the thermal resistance for air-cooled heatsinks (R_{SA}) is presented in references [7,8,9]. Beyond the sink thermal plane (T_s node) a generalized thermal solution (single-phase water-cooling, two-

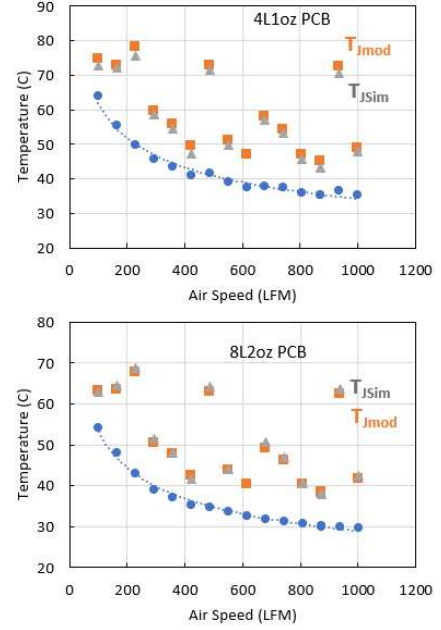


Fig. 5. Results verification for junction temperature for FET area range from 2.5 to 14 mm². Error in temperature rise within 10%

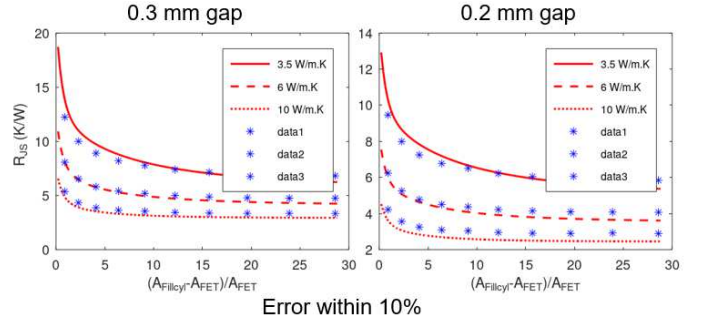


Fig. 6. Verification results for junction to sink resistance as function of TIM gap fill area and thermal conductivity.

phase cooling, radiation, ...) can be defined by a thermal resistance value (R_{SA}) to ambient temperature or another temperature reference (such as coolant supply temperature).

C. Resistive Circuit Model

After setting up the thermal resistances that constitute the thermal system, the circuit model would resemble the schematic shown in Fig. 3(a) for a single FET, and Fig. 3(b) for a half-bridge configuration. The full circuit is simplified to extract the overall resistances to ambient through the board-side path ($R_{th,J-PCB-A}$), and through the case-side path ($R_{th,J-C-A}$) if a back-side cooling solution is present. The model is built with a GUI for easy parameter input and setup, and the overall resistance to ambient is presented with the temperature rise estimation at the junction, along with all the resistance contributions of the individual components.

III. RESULTS

A. Verification with Numerical Simulations

The reduced model results are verified against numerical thermal simulations for different cases for the different sub-models. Initially, the board-side is verified for $R_{th,J-PCB}$ and $R_{th,PCB-A}$. The case-side model for the FETs are verified for different TIM materials with and without gap filler. The simulation models are built in 6SigmaET. The temperature is extracted from the junction and from a reference thermal plane (board or sink) to calculate the required thermal resistances.

1) PCB-to-ambient

The thermal resistance from junction to PCB is verified with numerical simulations by comparing the board temperature as function of air flow and board construction (stack-up and area). The temperature of the board is largely dependent on the resistance to ambient and thus is a weak function of FET parameters and vias placement. The graphs of Fig. 4 show that the model presents good estimation of the simulated thermal resistances for the different PCB constructions and air flow conditions.

2) Junction-to-PCB

The local temperature rise between the junction and the PCB region under the FET depends on the FET construction, board construction and on the placement of vias. The junction temperature and the board average temperature (PCB top and bottom) are used to calculate $R_{th,J-PCB}$ which is compared to the results obtained from thermal simulations. The results are reported for different board constructions (4 layers to 8 layers) and for FET die areas between 2.5 and 14 mm². Results in Fig. 5 show good agreement at the junction temperatures with less than 10% error.

3) Junction-to-Sink

The back-side thermal resistive circuit consists of the thermal resistances that the FET encounters against heat dissipation to the sink/cooling solution. The resistance of a TIM pad is mostly one-dimensional and can be accurately estimated from the (4). However, the added gap fill introduces addition thermal conduction paths that would reduce the overall thermal resistance to sink ($R_{th,JS}$). The results of Fig. 6 show good agreement between simulation results and the reduced model for $R_{th,JS}$.

4) Half-Bridge Configuration

To verify the half-bridge configuration, two identical FETs are placed with 2 mm spacing. In a set of cases, backside cooling with a 2mm heat spreader and an effective heat transfer of 250 W.m²K from its top surface is defined. TIM pad of thickness 0.3 mm is added atop the FETs. Thermovias are also defined for a subset of cases, considering that heat still conducts through the PCB. Comparing the model results to simulations show a good agreement with estimates

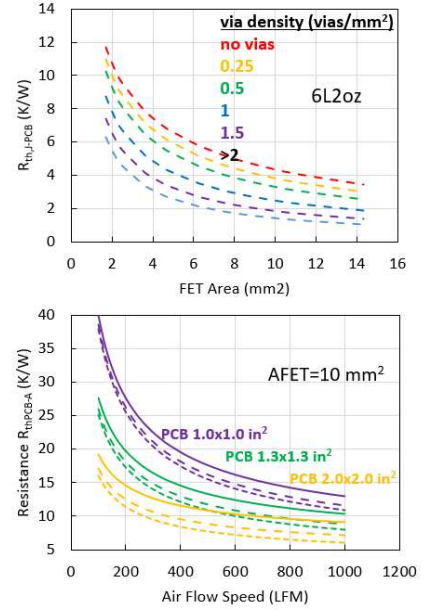


Fig. 7. Populated charts for thermal resistance of junction to board and board-to-ambient for different stack-up and PCB parameters

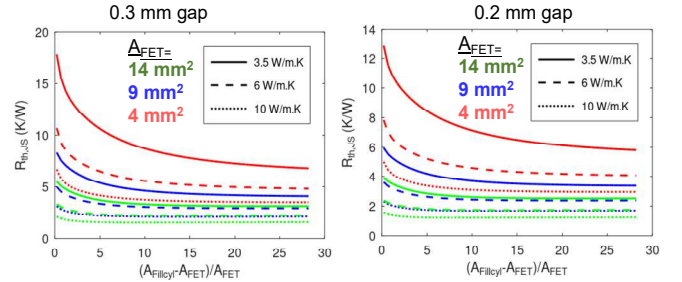


Fig. 8. Populated charts for junction to sink thermal resistance for different die areas and TIM thermal conductivity

within a 10% error margin (Table I).

B. Populated Look-up Charts

Using the model presented in this paper, look-up tables/charts can be generated to provide quick and convenient estimation tool for thermal resistances of CSP GaN FETs. The different charts provide close estimates for configurations including and excluding a backside cooling solution, using simple parameter inputs.

The resistance network to ambient through the PCB is simplified into two thermal resistances (Fig. 7), one representing the resistance from junction to the PCB ($R_{th,J-PCB}$), and another for the resistance from PCB to ambient ($R_{th,PCB-A}$). The overall resistance can be estimated by adding the two resistances.

TABLE I. RESULTS OF NUMERICAL VERIFICATION FOR SINGLE FET AND HALF BRIDGE CONFIGURATION, WITH AND WITHOUT BACK-SIDE COOLING (2X2IN PCB, 8L2OZ, IN 500 LFM, P1=5W, P2=2W, 2.5X3.6MM FET)

Half-Bridge	Back-side cooling ($h=250$ W/m ² K)	Vias/mm ²	T_J Model (°C)	T_J Sim. (°C)	Err (%)
No	No	0	78.5	88.3	10
No	No	1.3	67.6	70.5	5
No	Yes	1.3	43.5	45.0	5
No	Yes	0	45.9	48.4	10
Yes	No	0	111, 93	102, 84	9
Yes	No	1.3, 1	83, 74	82, 71	1
Yes	Yes	1.3, 1	55, 50	50, 46	10

Sample lookup graphs (Fig. 7) show the variation of the PCB thermal resistances as function of governing PCB and FET parameters. $R_{th,J-PCB}$ represents the resistance between the junction to the PCB and is thus localized near the FET region and is more sensitive to FET area and to the number of vias available under the FET (Fig. 7a). More vias improve conductance into the board, and thus reduce the resistance to heat conduction (Fig. 2a). The resistance $R_{th,PCB-A}$ represents the heat dissipation from the PCB into ambient and is less sensitive to local FET parameters. Air flow speed, and PCB's area are the major parameters affecting heat dissipation to ambient, as well as the effective thermal properties of the PCB, which in turn are a function of the number of conductor layers count and thickness (Fig. 7b).

The resistance to the back-side cooling solution is mostly governed by the thermal interface material. In addition to junction to case (R_{JC}), the initial thickness of TIM presents some resistance (R_{TIM}) which increases for smaller die areas. Assuming the same TIM above and around the FET, $R_{th,JS}$ can be reduced noticeably using gap filler, with the thermal benefit being more pronounced for smaller die sizes, and reducing $R_{th,JS}$ by up to 50%.

C. Experimental Validation

Considering the large number of independent parameters, extensive experimental validation would require a large set of test boards which would not be feasible. For that reason, a representative set of tests are performed on evaluation boards to validate the thermal simulations and the reduced thermal model. The tests will validate the junction-to-board thermal resistance (no back-side cooling), and with backside cooling for a single FET and two FETs in a half-bridge configuration. The experimental validation for junction to sink case was performed for smaller size die (EPC2204 FET 3.92mm² 2.8x1.4mm) and a larger size die (EPC2069 FET 10.56 mm² 3.25x3.25mm). Thermal testing of the activated PCBs is performed in flow-controlled wind tunnel. The air flow speed and ambient temperature is measured using degreeC® f200 temperature-compensated air flow sensor.

The experimental validation for $R_{th,JS}$ was presented in a previous study [5]. We compare those results to the estimates

TABLE II. COMPARISON OF JUNCTION TO SINK RESISTANCE MEASUREMENTS TO MODEL AND SIMULATIONS

κ_{TIM} (W/m.K)	R_{TIM+GF} (K/W) 0.3mm	R_{TIM+GF} (K/W) 0.2mm	Sim. (K/W) 0.3mm	Sim. (K/W) 0.2mm	Meas. (K/W)
3.5	6.8	5.9	6.7	5.8	6.1-6.2
6	4.6	3.9	4.8	4.1	4.6-5.1
10	3.1	2.6	3.4	2.9	-

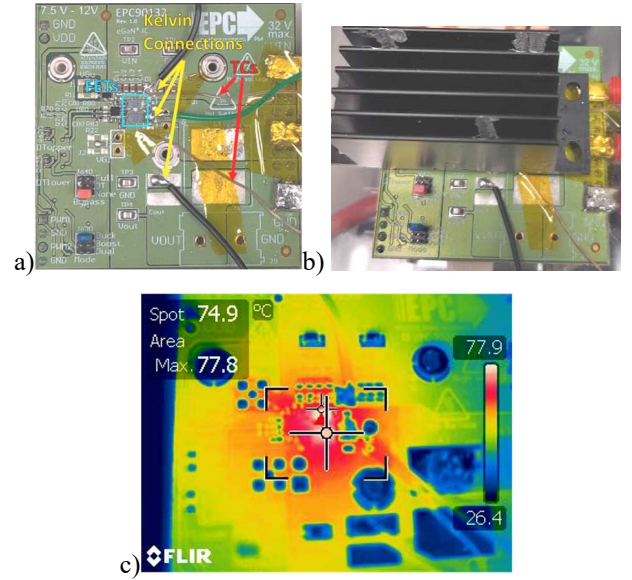


Fig. 9. a) PCB image showing FETs, TCs and Kelvin connections, b) PCB with mounted heatsink, c) IR image showing self heating of FETs (case: EPC2204, half-bridge in 200 LFM).

obtained by the reduced model (Table II) for different TIM conductivity (κ) showing close agreement. More details of the experimental setup for junction-to-sink thermal resistance measurement is presented in [5].

The first set of thermal tests in this work was performed with no backside cooling in forced air conditions between 200 and 600 LFM. The FETs are activated in reverse-bias with four-terminal connections to accurately set the power loss. Current was injected between the ground (GND) and switch node (SW) to bias the low-side FET, and between SW and V_{in} to bias the high-side FET. To activate both FETs (i.e., half-bridge), current was injected between V_{in} and GND. The temperature was measured using IR camera with a black painted die to increase emissivity. The board connections and a sample IR image is presented in Fig. 9. The results in Table III show that the measurements well validate the model with error within 15%, the only observation is that the measurement of the HB gives a higher difference of temperature between the FETs, and is most likely since the copper break between the FETs is not modeled.

The next set of measurements was performed with an added copper heat spreader and a 1/8th brick heatsink (Wakefield-Vette 567-24AB). SMD spacers provide a 0.3 gap

TABLE III. COMPARISON OF JUNCTION TO SINK RESISTANCE MEASUREMENTS TO REDUCED MODEL AND SIMULATIONS

	<i>EPC2204 (3.75mm²) High-side FET</i>			<i>EPC2204 (3.75mm²) Low-side FET</i>			<i>Mod. T</i>	<i>% err</i>	<i>EPC2069 (10.6mm²) High-side FET</i>			<i>EPC2069 (10.6mm²) Low-side FET</i>			<i>Model T</i>	<i>% err</i>
Air Flow (LFM)	P (W)	T (°C)	Rth (K/W)	P (W)	T (°C)	Rth (K/W)	T (°C)		P (W)	T (°C)	Rth (K/W)	P (W)	T (°C)	Rth (K/W)	T (°C)	
200	2.09	70	22.5	-	-	-	68.6	2	1.51	58	23.2	-	-	-	51.7	11
400	2.10	63	19.1	-	-	-	57.8	8	1.44	49.8	18.7	-	-	-	44.3	11
600	2.09	61.1	18.2	-	-	-	53.3	13	1.42	47.1	17.0	-	-	-	41.2	13
200	-	-	-	2.25	70.4	21.1	73.8	5	-	-	-	2.91	74.3	17.7	82.7	11
400	-	-	-	2.26	61.8	17.2	62.1	1	-	-	-	2.89	63.5	14.0	67.6	6
600	-	-	-	2.25	58.9	15.9	57.3	3	-	-	-	2.92	62.6	13.5	61.3	2
200	1.65	75.5	31.8	1.48	78	37.1	82.7, 85.7	10	1.85	87	30.9	1.80	82.5	28.3	90.1, 90.0	9
400	1.43	59	25.3	1.29	61	29.6	55.0, 56.5	7	1.85	72	26.5	1.80	68	25.0	71.5, 71.6	5
600	1.28	51	22.0	1.15	53	26.1	51.8, 52.0	2	1.85	64	22.2	1.80	60	20.5	63.4, 63.7	6

TABLE IV. THERMAL TESTING RESULTS WITH BACKSIDE-COOLING

	<i>EPC 2069 (10.6 mm²)</i>							<i>EPC2204 (3.75 mm²)</i>						
Air Flow (LFM)	P ₁ (W)	T ₁ (°C)	P ₂ (W)	T ₂ (°C)	Model T ₁ (°C)	T ₂ (°C)	Error (%)	P ₁ (W)	T ₁ (°C)	P ₂ (W)	T ₂ (°C)	Model T ₁ (°C)	T ₂ (°C)	Error (%)
200	3.19	61.3	3.33	58.2	59.02	58.77	4% 1%	4.02	93	4.52	92.8	96.66	98.45	4% 6%
400	3.20	47.8	3.33	44.5	45.09	44.84	6% 1%	4.12	68.1	4.49	67.9	64.44	62.68	5% 8%
600	3.21	43.8	3.35	40.6	42.95	42.7	2% 5%	4.19	63.3	4.53	62.8	60.93	59.19	4% 6%

for the TIM between FETs and the heat spreader. High conductivity T-Global A1780 TIM ($\kappa=17.8$ W/m.K) was used for the FETs TIM pads and A1660 TIM ($\kappa=16.6$ W/m.K) was used between heatsink and heatspreader. The heatsink was hard mounted using M2 screws to the SMD threader spacers. Temperature readings are obtained using fine gauge thermocouples (40 AWG) in contact with the FETs. The results presented in Table IV show that the model closely estimates the actual test results to within 10% error.

We present here the results in Table II showing a closely agreement between measured values and model results. Validation errors could arise due to non-ideal conditions in the test chamber, obstructions to air flow due to the components and wiring on the PCB, and the discontinuous copper layers within the PCB. The experimental testing results however show satisfactory accuracy in predicting and estimating thermal performance of GaN FETs.

IV. CONCLUSIONS

A reduced thermal model for CSP GaN FETs is presented to simplify the thermal system and provide quick estimates of thermal performance based on general build parameters. The model is implemented in a user-interface and also used to generate look-up graphs for simple PCB constructions and back-side cooling setups to provide a quick lookup approach to estimating the thermal performance (Fig. 7, Fig. 8) and the comparative cooling effectiveness of different parameters at the board level and at the backside. The presented models are verified against numerical simulations and validated with

experimental thermal testing and is seen to provide good accuracy. The effects of PCB size, stack-up, via placement, and selected TIM material are also described and discussed. The model is expanded to consider the thermal characteristics of a bridge configuration with two adjacent active FETs which is also beneficial for widely-used electronic configurations.

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